

Rifqi Afta Ronanda

AI and Data Engineering

Turning data into useful decisions.

I connect engineering thinking, machine learning, and clear interfaces to build practical tools for research and operations.

Engineering Physics background at Universitas Gadjah Mada. Exploring IT and AI engineering opportunities with companies in Japan.

FOCUS

Practical data workflows, machine learning, responsive web applications, and clear technical communication.

SELECTED WORK

Solar operations monitor | Local AI evidence database | Soundscape analysis pipeline

HOW TO READ THIS PORTFOLIO

Each case separates the implemented workflow from its intended business value and remaining validation. Commercial outcomes, production deployments, and performance gains are not claimed without evidence.

GitHub / [rifqironanda](#)

LinkedIn / [Rifqi Afta Ronanda](#)

Repository documentation reviewed: 7 October 2026. LinkedIn is provided as a profile link; detailed employment information has not been independently reviewed for this edition.

01 / Solar operations monitor

Research prototype | Python / Streamlit / Isolation Forest

PROBLEM

Low solar output can come from weather or unusual operating conditions. Teams need context before deciding what to inspect.

IMPLEMENTATION

A CSV-to-dashboard workflow combining data validation, an irradiance and temperature baseline, Isolation Forest per inverter, and explainable alerts.

INTENDED BUSINESS VALUE

Designed to help operations teams prioritise inspections and interpret underperformance with environmental context.

EVIDENCE

Repository documents CSV validation, inverter filtering, actual-versus-expected charts, and alert-logic unit tests.

VALIDATION NEEDED

No measured downtime reduction or verified fault-detection accuracy. Ambient temperature is a proxy; plant-specific calibration and maintenance records are needed.

[View project documentation on GitHub](#)

02 / Local AI evidence database

Local MVP | Python / Ollama / SQLite

PROBLEM

Literature review involves repetitive PDF extraction, summarisation, and organising evidence for later review.

IMPLEMENTATION

Local PDF discovery and page-text extraction with PyMuPDF, Ollama summaries, SQLite storage, and a manual validation state.

INTENDED BUSINESS VALUE

Designed to reduce repetitive document handling while keeping processing local and human review explicit.

EVIDENCE

Documented architecture separates documents, page text, summaries, and validations. No cloud LLM API is required by the default pipeline.

VALIDATION NEEDED

No measured time savings. Reliable claim extraction and citation verification remain roadmap items; summaries require human validation.

[View project documentation on GitHub](#)

04 / Soundscape analysis pipeline

Research pipeline | Python / MFCC / PCA / KMeans

PROBLEM

Audio collections need consistent representations before similarities and unusual recordings can be explored.

IMPLEMENTATION

MFCC, spectral and temporal features; statistical aggregation; PCA; KMeans clustering; and Isolation Forest outlier detection.

INTENDED BUSINESS VALUE

Creates reusable features, cluster labels, and outlier indicators to support exploratory review of environmental recordings.

EVIDENCE

Repository specifies feature, label, and outlier array outputs with a repeatable Python workflow.

VALIDATION NEEDED

Clusters are not verified species or semantic labels. No demonstrated biodiversity metric or commercial monitoring outcome.

[View project documentation on GitHub](#)